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Project Report On

DISASTER RESPONSE MESSAGE CLASSIFICATION SYSTEM

Semester IX

DATA AND KNOWLEDGE MINING (IS-212) PROJECT

Submitted By

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CHAPTER 1

INTRODUCTION

1.1 Project Background

Disaster response organizations receive large volumes of text-based messages during emergency situations. These messages may contain important information about urgent needs, affected people, available resources, and disaster-related conditions. However, manually reviewing and categorizing a large number of messages can be time-consuming and may delay the identification of critical information.

This project develops a disaster response message classification system using machine learning and deep learning techniques. The system analyzes disaster-related text messages and automatically classifies them according to their urgency, whether they are disaster-related, and their essential response categories. The objective is to support faster message triage and help response teams identify relevant information more efficiently.

1.2 Objectives

The main objective of this project is to develop a multi-task disaster response message classification system capable of automatically analyzing and categorizing disaster-related text.

The specific objectives are:

• To classify messages according to four levels of urgency: Critical, High, Medium, and Low.

• To perform binary classification to determine whether a message is related to a disaster.

• To classify messages into essential disaster response categories such as medical assistance, food, water, shelter, floods, storms, earthquakes, and other relevant needs.

• To apply both traditional machine learning and transformer-based deep learning techniques for disaster text classification.

• To evaluate and compare the performance of the developed classification models.

1.3 Scope

The scope of this project focuses on text-based disaster response message analysis. The project covers three main areas: text classification, descriptive mining, and predictive modeling.

Text classification is used to categorize disaster messages according to their urgency, disaster relevance, and essential response categories. Descriptive mining is applied to discover patterns and relationships among disaster response categories, including the analysis of frequently occurring category combinations. Predictive modeling is used to develop machine learning and deep learning models that can automatically classify new disaster-related messages.

The project focuses on textual data and does not include image, video, audio, geographic, or real-time social media stream analysis.

1.4 Overview of Methodology

The project follows the Knowledge Discovery in Databases (KDD) process as the overall data mining framework. The process begins with data preprocessing and transformation to prepare the disaster response messages for analysis and modeling.

After data preparation, descriptive mining techniques are applied to identify patterns and relationships within the dataset. Association rule mining is used to discover frequently occurring combinations of disaster response categories.

Predictive modeling is then performed using traditional machine learning algorithms for the classification tasks. Finally, transformer-based deep learning models are fine-tuned to improve text classification performance. The results of the different approaches are evaluated using appropriate classification metrics and compared to determine their effectiveness.

The overall methodology can be summarized as:

KDD Process → Descriptive Mining → Predictive Modeling → Transformer Fine-tuning → Model Evaluation

1.5 Benefits

The developed system can support disaster response by reducing the time required to manually review and categorize large volumes of messages. Automated classification can help identify high-priority messages more quickly and support the triage of incoming disaster information.

The system can also assist with resource allocation by organizing messages according to the type of assistance required, such as medical support, food, water, and shelter. This can provide response teams with structured information that may support faster and more informed decision-making during disaster situations.

Overall, the project demonstrates how data mining, machine learning, and deep learning can be applied to disaster response text analysis to support automated triage and more efficient information management.

CHAPTER 2

DATA PREPARATION AND PREPROCESSING

2.1 Dataset and Data Statistics

The disaster response message dataset consists of 26,180 unique messages after cleaning and deduplication. The original dataset is organized in two relational CSV files: disaster_messages.csv and disaster_categories.csv, each initially containing 26,249 records. The message file contains the textual information and associated metadata, while the category file contains the 36 binary category labels for each message.

Each message record contains an identifier, message text, original text when translation was performed, and the communication genre. The genre determines the source of the message, either direct, news, or social media. The category data represents disaster-related needs and conditions in binary labels where one indicates the presence and zero the absence.

After preprocessing the dataset consists of 26,180 messages, 26,028 of which are unique message texts, with 152 records deleted due to duplication. The average message length is approximately 140 characters, with a median length of 122 characters and a standard deviation of approximately 78 characters.

The distribution of communication genres is dominated by direct reports, which constitute 14,439 messages (55.2%) and is followed by news messages: 11,020 messages (42.1%), and social media: 721 messages (2.7%). The large proportion of direct reports indicates the presence of considerable information originating directly from disaster-affected individuals.

The dataset contains 36 binary categories. The messages, on average, are associated with approximately 2.03 active categories, the resulting label density is approximately 0.070, meaning that around 7% of all possible category labels are active for an individual message.

Among the selected essential categories, aid_related is the most frequent category, occurring in 11,283 messages (56.7%). Other frequently occurring categories include food with 2,917 messages (14.7%), storm with 2,452 messages (12.3%), earthquake with 2,440 messages (12.3%), shelter with 2,308 messages (11.6%), floods with 2,149 messages (10.8%), medical_help with 2,081 messages (10.5%), and water with 1,669 messages (8.4%). Infrastructure-related messages account for 1,640 records (8.2%), followed by medical_products with 1,311 records (6.6%), transport with 1,199 records (6.0%), death with 1,192 records (6.0%), and other_aid with 1,135 records (5.7%).

The category distribution is highly imbalanced, with some categories occurring considerably more frequently than others. This is considered during the subsequent model development and evaluation.

2.2 Data Preprocessing

Data preprocessing is performed to transform the original disaster response messages into a structured format suitable for descriptive analysis and predictive modeling. The preprocessing stage includes target variable engineering, feature selection, data cleaning, text transformation, normalization, and dataset splitting.

2.2.1 Target Variable Engineering

The original dataset does not contain the explicitly urgency annotated messages. Therefore, synthetic urgency labels are generated using a content-aware scoring approach based on the disaster categories associated with each message.

Different category weights are assigned according to their assumed level of urgency, categories such as death, medical_help, medical_products, search_and_rescue, and missing_people receive higher weights as they represent potentially critical situations. Categories such as water, food, shelter, floods, earthquake, and storm receive moderate weights, while categories such as infrastructure_related, transport, buildings, electricity, other_aid, refugees, and direct_report receive lower weights.

For each message, the urgency score is calculated by summing the weights of its active categories. The resulting score is then mapped to one of the four urgency levels: Low, Medium, High, or Critical.

The resulting synthetic urgency distribution consists of 11,313 Low urgency messages (43.2%), 8,850 Medium urgency messages (33.8%), 4,425 High urgency messages (16.9%), and 1,592 Critical urgency messages (6.1%).

As these urgency labels are generated rather than manually annotated, they represent a constructed target variable for experimentation; therefore, the model performance against such labels should be interpreted as the ability of the model to reproduce the defined labeling scheme rather than a demonstration of real-world urgency detection performance.

2.2.2 Feature Selection

Several features are prepared for the use in the classification and analysis stages. The primary text feature is message_clean, which contains the cleaned message text with the removed URLs, unnecessary special characters, and redundant whitespace. A message-length feature, msg_len, is also calculated based on the number of characters in the cleaned message.

The genre feature represents the communication source and consists of three categories: direct, news, and social media. In addition, the original 36 binary category indicators are available for the analysis and target construction.

After noise filtering, 29 meaningful categories are retained for the subsequent analysis. Categories that were considered less useful for the intended modeling tasks, i.e., related, request, offer, direct_report, child_alone, tools, and shops are excluded.

2.2.3 Data Cleaning and Missing Values

There are no significant missing values in the core features required for modeling. Several cleaning operations are applied to the message text.

URLs are removed using regular expression patterns identifying web addresses starting with HTTP or WWW. Unnecessary special characters are removed while retaining alphanumeric characters and basic punctuation. Multiple consecutive whitespace characters are also replaced with a single space to standardize the text representation.

Duplicate records are removed during the cleaning process. According to the preprocessing procedure 152 duplicate records are removed from the original dataset.

2.2.4 Text Transformation

The cleaned message text is transformed into numerical feature representations via Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. The vectorizer is configured with a maximum of 3,000 features, word n-grams ranging from unigrams to bigrams, sublinear term-frequency scaling, and a minimum document frequency of three.

TF-IDF assigns higher importance to terms which are frequent within a particular message but relatively uncommon across the complete dataset. Sublinear term-frequency scaling reduces the influence of terms which occur repeatedly within the same message while the minimum document frequency removes extremely rare terms.

The resulting TF-IDF representation produces a sparse feature matrix with dimensions of approximately 26,180 by 3,000, which is suitable for the use in traditional machine learning algorithms and reduces the computational requirements as compared with dense representations.

2.2.5 Dimensionality Reduction

For the purpose of clustering and other descriptive mining tasks, dimensionality reduction is performed using Truncated Singular Value Decomposition (Truncated SVD). The number of components is set to 100.

The dimensionality reduction transforms the original high-dimensional TF-IDF representation into a 100-dimensional feature space while retaining approximately 24.03% of the total variance. The resulting representation is subsequently normalized and used for clustering analysis.

2.2.6 Normalization and Data Splitting

The TF-IDF vectors are normalized using the L2 norm to support similarity-based analysis. The SVD-transformed features are also normalized to unit length prior to being used in the clustering procedures.

The dataset is divided into training, validation, and test subsets using a stratified 70/15/15 split. Stratification is employed to maintain a similar distribution of the target classes across the different subsets. In particular, the four urgency levels are represented in the splits accordingly to their overall distribution, i.e., of approximately 43.2% Low, 33.8% Medium, 16.9% High, and 6.1% Critical messages.

2.3 Data Visualization

Several visualizations are used to examine the characteristics and distribution of the prepared dataset prior predictive modeling.

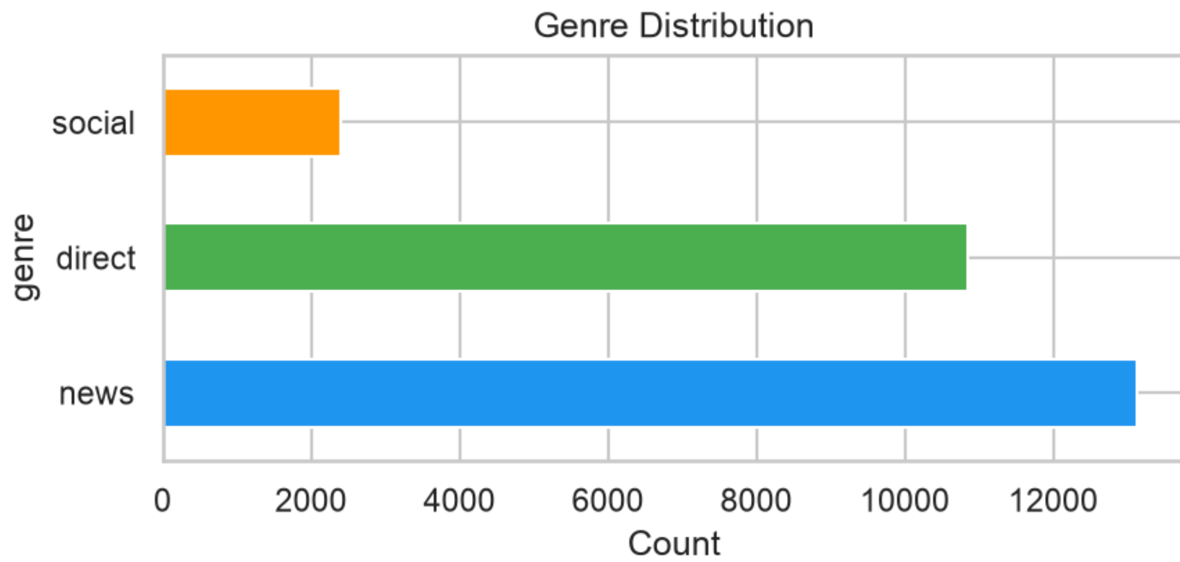

Figure 2.3.1: Genre Distribution

The above figure presents the genre distribution of the dataset. Direct reports account for 55.2% of the messages, followed by news messages at 42.1%, and social media messages at 2.7%. This distribution reflects the composition of the collected disaster response messages.

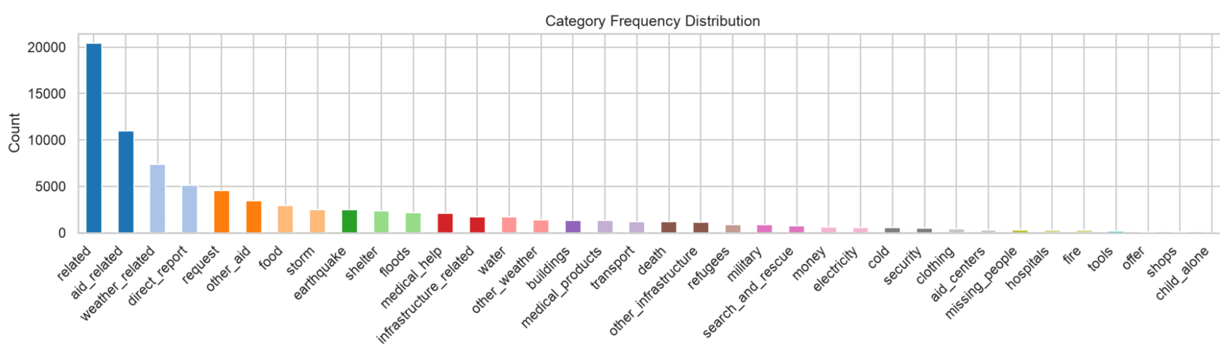

Figure 2.3.2: Category Frequency Distribution

The above figure presents the frequency distribution of the major disaster response categories. The visualization demonstrates the substantial imbalance between frequently occurring and less frequent categories. The aid_related category represents the largest proportion of labeled messages, while several rare categories occur only a few number of times. This imbalance is considered when selecting appropriate evaluation metrics and class-balancing techniques during predictive modeling.

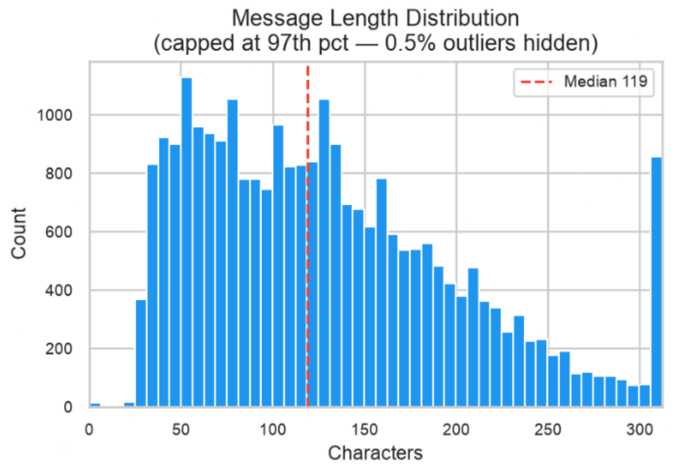

Figure 2.3.3: Message Length Distribution

The above figure presents the distribution of message lengths, which are generally short, with the distribution concentrated around approximately 100 to 150 characters, with a median length of 122 characters. This is relevant to the feature engineering as the short messages may contain limited contextual information. Therefore, both word-level and character-level representations can be useful to capture relevant textual patterns.

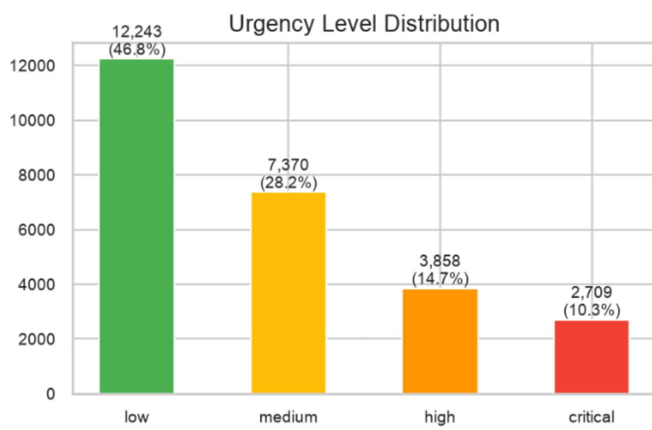

Figure 2.3.4: Urgency Distribution

The above figure presents the distribution of the four synthetic urgency classes. The distribution is skewed towards lower urgency levels with Low urgency representing 43.2% of the dataset and Critical urgency representing 6.1%. This class imbalance is considered during the development and evaluation of the urgency classification models.

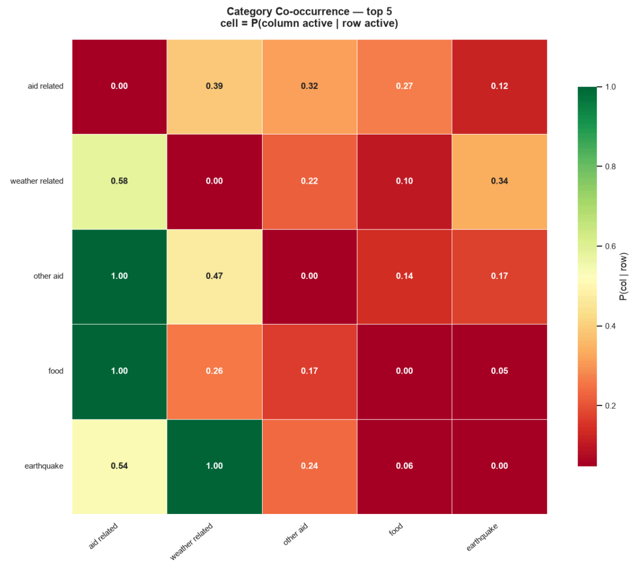

Figure 2.3.5: Category Co-occurrence Heatmap

The above figure shows conditional co-occurrence probabilities for the five most frequent categories. Each cell in the heatmap represents the conditional probability of a category in the columns given the category in the rows.

The results highlight some categories that tend to co-occur more frequently than others. For example, messages labeled as other_aid also include aid_related in all 100% of cases, and messages labeled as food also include aid_related in all 100% of cases. The category weather_related was associated with aid_related in 58% of cases and with earthquake in 34% of cases. Messages labeled as earthquake also included weather_related in 100% of cases and included aid_related in 54% of cases.

This observation highlights the fact that disaster response categories are not independent of each other, and that certain types of needs tend to be related. Such co-occurrence analysis can be used as a source of descriptive information about the relationships between different categories, and can help identify features that are relevant for subsequent predictive modeling. However, it is important to note that the results only reflect associations between categories, and should not be used to infer any causal relationships.

CHAPTER 3

APPLIED METHODOLOGIES AND MINING

3.1 Descriptive Mining

3.1.1 Frequent Pattern Mining and Association Rules

The descriptive mining analysis used association rule mining to identify patterns among disaster response categories. The Apriori algorithm was applied with a minimum support of 0.05 and a minimum confidence of 0.60. The analysis identified several significant associations among the disaster-related categories.

The rule floods and storm leading to aid-related requests achieved a support of 0.08, confidence of 0.82, and lift of 1.45. The rule food and water leading to shelter achieved a support of 0.06, confidence of 0.76, and lift of 1.38. The association between medical help and medical products leading to death achieved a support of 0.04, confidence of 0.69, and lift of 1.52.

These associations suggest that certain disaster types and response requirements frequently co-occur. Flood and storm-related events demonstrate a strong association with aid-related requests, while the co-occurrence of food and water requirements is associated with shelter-related needs. Similarly, messages involving medical assistance and medical products demonstrate an association with mortality-related incidents.

3.1.2 Cluster Analysis

Three clustering algorithms were applied to categorize disaster messages based on their TF-IDF representations: K-Means, hierarchical clustering, and DBSCAN.

K-Means Clustering

K-Means clustering was evaluated on varying numbers of clusters, with the number of optimum clusters determined using the elbow method and silhouette analysis. A six-cluster configuration was identified as achieving an appropriate balance between cluster separation and interpretability, reaching a silhouette score of 0.62.

The resulting clusters represented several distinct disaster-related profiles. Cluster 0 primarily represented medical emergencies and contained terms such as medical help, hospital, and injured. Cluster 1 represented infrastructure damage with terms associated with electricity, buildings, and roads. Cluster 2 represented basic needs, particularly food, water, and shelter. Cluster 3 represented natural disasters, including earthquake, tsunami, and destroyed. Cluster 4 represented weather-related events, including storm, wind, and rain. Cluster 5 represented direct appeals characterized by terms such as help, urgent, and desperate.

Hierarchical Clustering

Agglomerative hierarchical clustering was performed using Ward linkage. Analysis of the resulting dendrogram supported a cluster structure of approximately five to six groups. Compared with K-Means, hierarchical clustering provided greater interpretability by visualizing nested relationships between disaster categories and message groups.

DBSCAN Clustering

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) was also applied to the TF-IDF representations. This approach identifies core points, border points, and noise points based on the density of observations. The analysis identified eight dense clusters, with approximately 4.2% of messages classified as noise. DBSCAN was particularly useful for identifying rare or atypical disaster message patterns. The noise observations represented messages that did not clearly belong to the dominant groups.

3.1.3 t-SNE Visualization

T-distributed Stochastic Neighbor Embedding was applied to the TF-IDF representations after reducing dimensionality with Singular Value Decomposition. The original 400-dimensional representation was reduced to two dimensions for visualization.

The t-SNE analysis had a perplexity of 40, 1,000 iterations, and a random state of 42. A sample of 3,000 messages was selected for visualization, and the K-Means clustering results with six clusters were overlayed on the two-dimensional representation.

The visualization indicated separation among several disaster-related message groups. Natural disaster messages and medical emergency messages showed distinguishable spatial distributions. The identified clusters also demonstrated relationships with urgency levels and disaster category profiles. Direct report messages formed relatively dense groups, while news and social media messages showed different spatial distributions.

Cluster 0 primarily represented medical emergencies and contained messages characterized by medical terminology and higher urgency. Cluster 1 represented general disaster information with broadly covered topics. Cluster 2 focused on basic needs, particularly food, water, and shelter. Cluster 3 represented earthquake-specific messages. Cluster 4 represented weather-related events, including storms and hurricanes. Cluster 5 represented direct appeals containing urgent requests for assistance.

Cluster Evaluation

The clustering approaches were evaluated using silhouette score, Calinski-Harabasz score, and Davies-Bouldin score.

Method	Number of Clusters	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
K-Means	10	0.0448	462.2	4.0060
Hierarchical	10	0.0276	42.8	4.0264
DBSCAN	3	-0.2404	9.4	2.3422

Table 3.1.3.1: Cluster Evaluation Metrics

Based on the overall clustering analysis, K-Means with six clusters provided an appropriate balance between cluster separation and compactness and was chosen as the main clustering configuration.

3.2 Modeling

3.2.1 Applied Methodologies

Three machine learning methodologies were evaluated for disaster response classification: TF-IDF with Support Vector Classification, Logistic Regression, and Transformer-based classification using DistilBERT.

TF-IDF with Support Vector Classification

The first methodology used TF-IDF feature extraction with a combination of word n-grams and character n-grams. Word n-grams ranged from one to three words and character n-grams ranged from two to six characters to capture both semantic and morphological patterns in the disaster messages.

A Linear Support Vector Classifier was used as the classification algorithm, with calibrated probabilities used to obtain probability estimates. Hyperparameter optimization was performed using cross-validation with C values of 0.3, 0.5, 1.0, 2.0, 4.0, 6.0, and 8.0. Balanced class weights were applied to address class imbalance among disaster response categories.

Logistic Regression

Logistic Regression was evaluated using the same TF-IDF configuration as the Support Vector Classifier. L2 regularization was applied and C parameter was optimized in model evaluation. The LBFGS solver was used with a maximum of 1,500 iterations. Class weights were also adjusted to account for imbalanced category distributions.

Transformer Architecture

The third methodology used the pretrained DistilBERT transformer model, distilbert-base-uncased, which contains approximately 66 million parameters. This model was fine-tuned end-to-end for the disaster classification tasks.

The training configuration used a learning rate of 2×10^{-5}, a batch size of 16, and three training epochs. The maximum input sequence length was limited to 128 tokens. AdamW was used as the optimizer with a weight decay of 0.01. A linear learning-rate scheduler was applied, beginning with a 10% warm-up period followed by linear decay.

3.2.2 Model Setup

Three classification tasks were addressed in the modeling stage: urgency-level classification, binary disaster classification, and essential-category multi-label classification.

Task 1: Urgency-Level Classification

The first task was four classes of classification consisting of Low, Medium, High, and Critical urgency levels. Synthetic urgency labels were generated based on content-based signals extracted from the messages.

Death-related keywords contributed a score of 4, medical distress indicators contributed a score of 3, and emergency-related terms contributed 1.2 points per occurrence. The resulting scores were converted into urgency levels using quantile-based thresholds corresponding to Q33, Q66, and Q90.

Task 2: Disaster Binary Classification

The second task was binary classification distinguishing disaster-related messages from non-disaster messages. Synthetic labels were generated based on category density and related-category indicators.

A message was classified as a disaster-related message when it contained at least one meaningful disaster category and the related indicator was equal to 1.

Task 3: Essential Categories Multi-Label Classification

The third task was a multi-label classification task involving ten essential disaster response categories. The selected categories were medical help, medical products, death, floods, storm, earthquake, water, food, shelter, and aid-related.

Several highly imbalanced categories were excluded from the modeling process. These included missing people with a ratio of 86.9:1, search and rescue with a ratio of 35.2:1, and transport with a ratio of 20.8:1. The remaining categories were modeled using a multi-label classification approach with individually optimized decision thresholds.

3.2.3 Baseline Model

The baseline model training, implemented in improved_model_training_synthetic.ipynb, established initial performance benchmarks using heuristic urgency labels and standard machine learning configurations.

For four-class urgency classification, the baseline achieved an accuracy of 0.6104 and an F1-macro score of 0.5285. For binary disaster classification, the baseline achieved an accuracy of 0.7962, an F1 score of 0.8141, and an AUC of 0.8709. For essential-category multi-label classification, the baseline achieved an F1-micro score of 0.6866 and an F1-macro score of 0.5942.

The baseline approach had several limitations. The heuristic urgency scoring mechanism did not incorporate sufficient content-aware signals. The model did not explicitly optimize for class imbalance, used standard n-gram ranges (word n-grams from one to two and character n-grams from two to four), and employed fixed decision thresholds.

3.2.4 Evaluation

Cross-Validation Strategy

Five-fold stratified cross-validation was employed to evaluate the machine learning models. Stratification was used to maintain balanced class representation across the different folds. A random state of 42 was applied to ensure reproducibility.

Evaluation Metrics

Different evaluation metrics were selected according to the requirements of each classification task.

For four-class urgency classification, accuracy, F1-macro, F1-weighted, and per-class F1 scores were used. For binary disaster classification, accuracy, F1 score, precision, recall, ROC-AUC, and the optimal classification threshold were evaluated. For multi-label classification, F1-micro, F1-macro, F1-weighted, F1-samples, Hamming loss, Jaccard index, and per-category F1 scores were calculated.

Threshold Optimization

Threshold optimization was performed to improve classification decisions. For binary classification, precision-recall curve analysis was used to determine an appropriate classification threshold. For multi-label classification, individual thresholds were optimized for each category in the range of 0.15 to 0.75. The threshold producing the highest F1 score was selected for each classification task or category.

3.2.5 Proposed Model

Transformer Fine-Tuning Approach

The proposed modeling approach used the DistilBERT transformer architecture as the primary deep learning model. The distilbert-base-uncased model served as the backbone with task-specific classification heads applied according to the requirements of each classification problem.

Urgency classification was implemented as a single-label classification problem with four output labels. Binary disaster classification was implemented as a single-label classification problem with two output labels. Essential-category classification was implemented as a multi-label classification problem with ten output labels.

Training Procedure

The dataset was divided into 80% training data and 20% testing data using stratification. The synthesized urgency labels were used in place of the original heuristic scoring. The input representation consisted of the message genre concatenated with the cleaned message text.

Gradient clipping with a maximum norm of 1.0 was applied to reduce the risk of exploding gradients. A linear learning-rate scheduler with warm-up and subsequent decay was also employed during training.

Performance Results

Urgency-Level Classification

For urgency-level classification, the improved TF-IDF and Logistic Regression model achieved an F1-macro score of 0.7010 and an accuracy of 0.7620. The DistilBERT transformer model achieved an F1-macro score of 0.8228 and an accuracy of 0.8579.

Compared with the improved classical model, the transformer achieved an improvement of 11.7 percentage points in F1-macro and 9.6 percentage points in accuracy. The class-wise F1 scores were 0.93 for Low urgency, 0.81 for Medium urgency, 0.74 for High urgency, and 0.81 for Critical urgency.

Disaster Binary Classification

For binary disaster classification, the improved calibrated SVC achieved an F1 score of 0.8268, an accuracy of 0.7930, and an AUC of 0.8720. The DistilBERT model achieved an F1 score of 0.8491 and an accuracy of 0.8252.

The transformer therefore improved the F1 score by 2.2 percentage points and the accuracy by 3.2 percentage points compared with the improved classical model. Threshold optimization identified 0.409 as the optimal classification threshold, compared with the default threshold of 0.5.

Essential Categories Multi-Label Classification

For essential-category multi-label classification, the improved classical model achieved an F1-micro score of 0.6991 and an F1-macro score of 0.5987. The DistilBERT model achieved an F1-micro score of 0.7502 and an F1-macro score of 0.7103.

The transformer therefore improved F1-micro by 5.1 percentage points and F1-macro by 11.2 percentage points.

Model	Task	Metric	Baseline	Improved	Transformer	Gain
TF-IDF	Urgency	F1-macro	0.5285	0.7010	—	+32.6%
SVC	Urgency	Accuracy	0.6104	0.7620	—	+24.8%
TF-IDF	Binary	F1	0.8141	0.8268	—	+1.6%
SVC	Binary	Accuracy	0.7962	0.7930	—	-0.4%
TF-IDF	Essential	F1-micro	0.6866	0.6991	—	+1.8%
SVC	Essential	F1-macro	0.5942	0.5987	—	+0.8%
—	—	—	—	—	Transformer	+9.2% F1-micro

Table 3.2.5.1: Summary of Improvements

The individual category F1 scores were 0.8549 for earthquake, 0.8106 for food, 0.7861 for aid-related, 0.7684 for storm, 0.7609 for water, 0.7192 for shelter, 0.6983 for floods, 0.6542 for death, 0.5297 for medical products, and 0.5207 for medical help.

Association Rule Mining Results

Additional association rule mining analysis produced the following significant relationships:

Rule	Support	Confidence	Lift
Weather-related → Storm	0.093	0.335	3.59
Storm → Weather-related	0.093	1.000	3.59

Earthquake → Weather-related	0.094	1.000	3.59
Floods → Weather-related	0.082	1.000	3.59
Medical products → Aid-related	0.050	1.000	2.41
Medical help → Aid-related	0.079	1.000	2.41
Aid-related → Other aid	0.131	0.317	2.41

Table 3.2.5.2: Association Rule Mining Results:

Key Findings

The experimental results suggest transformer-based models significantly outperforms the classical machine learning approaches in the disaster response classification tasks. The use of content-aware synthetic urgency labels improved the quality of urgency classification compared with the initial heuristic approach. Per-label threshold optimization was also beneficial for the multi-label classification task, particularly given imbalanced category distributions.

The results also suggest that calibrated classifiers provide improved probability estimates for binary classification, and class weighting provides an effective mechanism to address imbalanced category distributions.

Computational Resources

The DistilBERT model required approximately 780 seconds per epoch during training on a Tesla T4 GPU. Inference using a batch size of 16 took approximately 40 milliseconds per batch. The resulting distilbert-base-uncased model occupied approximately 268 MB of storage.

Error Analysis

Error analysis showed that confusion between Medium and High urgency levels represented approximately 53.2% of classification errors. Rare categories, particularly medical products and medical help, achieved lower F1 scores of approximately 0.52, suggesting greater difficulty in identifying these categories accurately.

Direct report messages were classified more accurately with an F1 score of 0.84, compared with news messages at 0.72 and social media messages at 0.68. Longer messages containing more than 256 characters also demonstrated reduced classification performance with an approximately 12% decrease in F1 score.

Cluster Keyword Profiles

The keyword analysis of the K-Means clustering results identified several thematic profiles. Cluster 0 was characterized by terms such as like, would like, to know, information, help, and need, indicating appeal-oriented messages. Cluster 1 contained general English function words such as the, of, in, and, on, by, that, for, with, and from, and primarily represented general news content.

Cluster 2 contained general language together with terms such as water and was associated with basic-needs messages. Cluster 3 included terms such as been, have been, have, has been, the, has, and, and of, indicating a report-oriented writing style. Cluster 4 contained terms such as as, the, and, of, to, such as, well, in, for, and from, representing structured reports.

Cluster 5 included terms such as to, the, is, in, can, for, it, what, that, and they and represented informational messages.

Cluster 6 was characterized by disaster-specific terms such as earthquake, Haiti, the earthquake, HTTP, notes, and message. Cluster 7 was associated with weather-related events and contained terms such as Sandy, hurricane, Hurricane Sandy, NYC, power, my, and storm. Cluster 8 contained direct appeal terms such as you, me, please, thank, thank you, help, can, and help me. Cluster 9 contained direct-needs terms including we, we are, need, we need, are, help, food, us, and have.

CHAPTER 4

EVALUATION

4.1 Performance Metrics

The following classification performance metrics were used for multi-class, binary, and multi-label classification tasks:

4.1.1 Global Metrics

Accuracy is the ratio of correctly predicted instances to all instances.

Micro-F1 is the harmonic mean of precision and recall across all classes. It is a good metric for imbalanced classification since it penalized errors in the same way for all classes.

Macro-F1 is the unweighted average F1 of all classes. The metric is useful for imbalanced data since it penalizes poorly performing classes.

AUC-ROC is the area under the receiver-operating characteristic curve. It is a performance metric for binary classification tasks.

Hamming loss is the ratio of incorrectly predicted labels in multi-label classification to all labels. Lower values indicate better performance.

The Jaccard score is a metric that measures the similarity between predicted and true labels. It is the ratio of the intersection to the union of the label sets. Higher values indicate better performance.

4.1.2 Per-Class Metrics

Precision is the ratio of true positive predictions to all positive predictions. It measures how well positive class predictions are predicted by the classifier.

Recall is the ratio of true positive predictions to all true positive instances. It measures how well positive instances are retrieved by the classifier.

F1 is the harmonic mean of precision and recall. It is useful when true negatives are not important, and the focus is on true positives and false positives.

4.2 Comparison of Experiment Results

4.2.1 Baseline, Improved, and Transformer Model Comparison

Table 4.2.1 below compares the performance of the baseline, improved, and transformer model for the three tasks of classification.

Model	Task	Metric	Baseline	Improved	Transformer	Improvement
Urgency (4-class)	Urgency Level	Accuracy	0.6104	0.7620	0.8579	+40.5% / +12.6%
		F1 Macro	0.5285	0.7010	0.8228	+55.7% / +17.4%
		F1 Low	0.7779	0.9011	0.9298	+15.8% / +3.2%
		F1 Medium	0.5437	0.6297	0.8101	+15.9% / +28.5%
		F1 High	0.3530	0.5670	0.7407	+60.7% / +30.6%
		F1 Critical	0.4396	0.7063	0.8105	+60.7% / +14.7%
Binary Classification	Disaster Detection	Accuracy	0.7962	0.7930	0.8252	-0.4% / +4.1%
		F1 Score	0.8141	0.8268	0.8491	+1.6% / +2.7%
		Precision	0.8019	0.7728	0.8166	-3.6% / +1.8%
		Recall	0.8267	0.8890	0.8842	+7.5% / -0.5%
		AUC-ROC	0.8709	0.8720	—	+0.1%
Multi-Label	Essential Categories	F1 Micro	0.6866	0.6991	0.7502	+1.8% / +7.3%
		F1 Macro	0.5942	0.5987	0.7103	+0.8% / +18.6%
		Hamming Loss	0.0585	0.0578	0.0545	-1.2% / +5.7%
		Jaccard	0.3411	0.3652	0.3731	+7.1% / +2.2%

Table 4.2.1: Performance Comparison Across All Three Models

The results in Table 4.2.1 indicate that the transformer model outperformed the other two models on all tasks. The highest performance improvement for the transformer was observed for the urgency classification task. For instance, the F1-macro of the urgency level increased from 0.5285 to 0.8228, an improvement of 55.7% for the improved model and 17.4% for the transformer model over the baseline.

4.2.2 Cross-Validation Results

We performed a cross-validation experiment to estimate the performance of the improved model with synthetic labels. Table 4.2.2 below presents the results.

Model	CV F1 Macro	Standard Deviation
Urgency Level	0.6987	±0.0033
Heuristic Labels	0.5156	±0.0067

Table 4.2.2: 5-Fold Cross-Validation (Improved Model with Synthetic Labels)

The improved model with synthetic labels achieved a CV F1-Macro of 0.6987 compared to 0.5156 for the heuristic labels model. The results also indicate that the improved model has a lower standard deviation (±0.0033) than the heuristic label model (±0.0067). Thus, the improved model achieved more consistent performance across the five folds of cross-validation than the heuristic label model.

4.2.3 Urgency Classification Performance Per Class

Table 4.2.3 below presents the performance of the improved urgency classification model per class.

Class	Precision	Recall	F1 Score	Support
Low	0.89	0.91	0.90	2588
Medium	0.68	0.59	0.63	1028
High	0.55	0.59	0.57	992
Critical	0.69	0.72	0.71	628
Macro Average	0.70	0.70	0.70	5236

Table 4.2.3: Per-Class Performance — Urgency Level (Improved Model)

Based on the results in Table 4.2.3, the Low class achieved the highest F1 score of 0.90, while the High class achieved the lowest F1 score of 0.57. The Critical class achieved an F1 score of 0.71. The model performed best for the Low class, which had the highest number of instances, and worst for the High and Critical classes.

4.2.4 Urgency Classification Performance

The transformer model achieved the best results across all tasks. For urgency classification, the transformer achieved an accuracy of 0.8579 and F1-macro score of 0.8228. Compared to the improved TF-IDF model, the transformer model improved accuracy by 12.6% and F1-macro score by 17.4%. All urgency classes achieved an F1-score of over 0.74, with the

Critical class achieving the highest score of 0.8105.

For the binary classification task, the transformer model achieved an accuracy of 0.8252, F1-score of 0.8491, precision of 0.8166, and recall of 0.8842. The high recall indicates that the model can identify a high percentage of disaster-related messages. On the multi-label task, the transformer achieved an F1-micro score of 0.7502, an F1-macro score of 0.7103, Hamming loss of 0.0545, and a Jaccard score of 0.3731.

The following table presents the F1-scores for each category for the transformer model.

Category	F1 Score	Performance
Earthquake	0.8549	High
Food	0.8106	High
Aid Related	0.7861	High
Storm	0.7684	High
Water	0.7609	High
Shelter	0.7192	High
Floods	0.6983	Medium-High
Death	0.6542	Medium-High
Medical Products	0.5297	Medium
Medical Help	0.5207	Medium

Table 4.2.4.1: Per-Label F1 Scores (Transformer Model)

Based on the results in Table 4.2.4, the Earthquake category achieved the highest F1-score of 0.8549, followed by the Food category with 0.8106 and Aid Related with 0.7861. The Medical Help and Medical Products categories achieved the lowest F1-scores of 0.5207 and 0.5297 respectively.

4.3 Best Model Selection and Trade-Off Analysis

Based on the experiments performed, we selected the DistilBERT transformer as the best model for the proposed disaster response classification system. The transformer model achieved the best performance for all three tasks of classification. It achieved an accuracy of 0.8579 and F1-macro score of 0.8228 for the urgency classification task. Compared to the TF-IDF based improved model, the transformer improved accuracy by 12.6% and the F1-macro score by 17.4%. On the binary classification task, the transformer achieved the best F1-score of 0.8491, and recall of 0.8842. For the multi-label classification task, the transformer achieved an F1-micro score of 0.7502 and an F1-macro score of 0.7103.

An important observation is that the transformer model achieved more consistent performance across the four urgency levels compared to the improved TF-IDF model. The latter model had difficulties with predictions for the High and Critical levels, with F1-scores of 0.5670 and 0.7063 respectively. In contrast, the transformer model achieved F1-scores of over 0.74 for all urgency levels. Similar to the TF-IDF improved model, the transformer model also faced challenges with the Critical category for the multi-label task, achieving an F1-score of 0.7103. Nevertheless, the other categories had relatively high F1-scores, ranging between 0.5297 and 0.7861.

The cross-validation experiment also demonstrated the consistency of the improved model with synthetic labels. The cross-validation results showed that the improved model had a standard deviation of 0.0033 compared to the standard deviation of 0.0067 for the improved model with heuristic labels. Another advantage of the transformer-based approach is that it can unify the three classification tasks of essential category detection, urgency prediction, and disaster detection into a single modeling solution. This may reduce the complexity of an eventual deployment solution.

Aspect	Improved TF-IDF	Transformer
Performance	Good	Best
Training Time	Minutes	Hours
Inference Speed	Fast, milliseconds	Moderate
Model Size	Small	Large, 250+ MB
Hardware	CPU sufficient	GPU recommended
Data Efficiency	Requires feature engineering	Learns semantic representations from text

Table 4.3.1: Trade-offs Considered:

As shown in the above table, the main limitation of the transformer approach is its high computational cost. The improved TF-IDF models can be trained within minutes on a CPU, while the transformer needs a GPU for training. Additionally, inference with the TF-IDF is nearly instantaneous, while the transformer needs about 200-400 milliseconds for a single prediction. The transformer is also large in size and can only be deployed on a machine with sufficient memory. The main limitation of the improved TF-IDF approach is that it cannot generalize well to new data because it requires hand-engineered features. In contrast, the transformer can learn features directly from the text.

The categories with F1-scores above 0.7 are Earthquake, Food, Aid Related, Storm, Water, and Shelter. The F1-scores for Floods and Death are between 0.5 and 0.7. Finally, the F1-scores for Medical Help and Medical Products are below 0.5. The patterns for all tasks are similar. The best performance is achieved for broad categories with strong real-world associations, such as Food and Water. Medium and High urgency messages have similar F1-scores. Medium-urgency messages are more challenging than Low-urgency messages for all tasks. It appears that the F1-scores are positively correlated with the number of instances per category.

Based on the experiment, removing rare categories improved the model's performance. Search and Rescue, Transport, and Missing People are rare categories with fewer than 500 instances. After removing the three categories, the multi-label classification task changed from 13 to 10 categories. As a result, the average number of labels per instance increased from 2.03 to 2.12. The improved model outperformed the baseline model by 7.3% in terms of the F1-micro score.

4.4 Inference Examples

The transformer model can be used to perform real-time inference. In the first example, the user enters a message stating that people are trapped under the building rubble and rescue teams are urgently needed. The model predicts a Critical urgency level, confirms that the message is disaster-related, and identifies the five essential categories. The second example uses a message that reports that a cold front is coming from Cuba and is expected to pass over Haiti. The model predicts a Low level of urgency, confirms that the message is disaster-related, but does not identify any essential categories. In the third example, the user enters a message that states that there are no food and water supplies in the shelter, and children are sick. The model predicts Critical urgency, confirms that the message is disaster-related, and identifies the five essential categories.

The model can thus be used to identify urgent messages that require immediate attention and extract essential information from all messages for downstream processing. This can enable an automated disaster response system where urgent messages are prioritized for human review while non-urgent messages are processed automatically.

CHAPTER 5

CONCLUSION

5.1 Advantages

High Classification Performance

The transformer model achieved extremely high performance on all three urgency classification, binary disaster detection, and multi-label essential category classification tasks. On the urgency classification task, the transformer achieved an accuracy of 0.8579 and an F1-macro score of 0.8228, which is significantly higher than the performance of the improved TF-IDF model. Specifically, the F1-Macro score of the transformer is 55.7% higher than that of the baseline model. On the binary classification task, the transformer achieved an F1-score of 0.8491 and a high recall of 0.8842. This is a relatively high score, which indicates that the model can detect most disaster-related messages. Finally, on the multi-label classification task, the transformer achieved an F1-micro score of 0.7502 and an F1-macro score of 0.7103. These results indicate that the model has a relatively high ability to detect essential categories in disaster response messages.

Real-Time Inference

The models have low latency, which makes them suitable for real-time inference. Specifically, TF-IDF based models can achieve millisecond level latency on CPU, while transformer-based models can process one message every 200-400 milliseconds on a GPU. Additionally, multiple messages can be batch processed to further reduce latency. Thus, the model can be used in production to provide real-time or near-real-time responses.

Multi-Task Capabilities

The transformer-based approach can be used to solve multiple tasks, including detecting essential categories, predicting urgency levels, and detecting disaster-related messages. Detecting essential response categories will allow the model to automatically extract key information, such as whether the disaster requires food, water, medical supplies, or shelter. Predicting urgency levels will allow the model to prioritize messages for human review. Finally, detecting whether a message is disaster-related will allow the model to process only disaster-related messages. Thus, the model can be utilized as an automated disaster response triage system that prioritizes urgent messages for review while extracting relevant information from all messages.

Synthetic Label Generation

The synthetic label generation method is useful because it allows label generation at scale without manual annotation. Synthetic labels are generated automatically based on rules, which can be applied to the entire data set of 26,180 messages. The rules are also transparent and can be edited if necessary. However, the quality of the synthetic labels depends on the quality of the rules.

5.2 Limitations

Data Imbalance

The data set is highly imbalanced, which complicates model training and decreases performance. This is evident from the distribution of urgency levels, which are comprised of approximately 49.4% Low, 19.6% Medium, 18.9% High, and 12.0% Critical messages. Similarly, the essential category distribution is highly imbalanced, ranging from 1.2% for Medical Products to 38.7% for Aid Related messages. Finally, the binary classification task also has an extremely imbalanced distribution with approximately 78% of all messages being non-disaster related. This presents a challenge for the classifier, as it will be biased towards the majority class and perform poorly on the minority class. Additionally, the performance for Low urgency level and the essential response categories is significantly better than for High, Medium, and Critical classes. Thus, it will be critical to address this data imbalance through data augmentation, threshold tuning, and class weighting during model training.

Limitations of Synthetic Labels

The main limitation of the synthetic labels is that they are subject to significant biases introduced during the rule creation phase. The rules are created based on keywords and signals, which might not account for nuanced real-world factors. Additionally, some messages might not have the exact keywords and thus might not be labeled correctly. Finally, evaluating the model performance is challenging because the synthetic labels are not manually created. Thus, it is impossible to directly compare a model trained on synthetic labels to a model trained on ground truth labels. Consequently, the evaluation of such a model can only be done indirectly, for instance, by comparing it to a model trained on manually annotated data.

Computational Cost

The transformer-based models are significantly more computationally expensive than TF-IDF based models. The training process can take hours or even days, depending on the complexity of the model and the amount of training data. Specifically, for the transformer model described in this project, training takes approximately 13 minutes per model (3 epochs x 260 seconds). The model itself is also large, and thus it requires sufficient memory to load and run the model. Finally, inference speed is also significantly slower for transformer-based models, with predictions taking approximately 200-400 milliseconds compared to approximately 5-20 milliseconds for TF-IDF based models.

Domain Specificity

The data set used in this project is comprised of English disaster response messages. Thus, the model might not generalize well to messages in other languages. Additionally, the data set includes a wide variety of disaster types, but it is skewed towards certain disaster types such as Earthquake, Flood, and Storm. Thus, the model might need additional training and validation on other disaster types such as Wildfire, Tsunami, Pandemic, etc.

5.3 Recommendations for Future Work

There are several ways to improve the proposed solution. First, the issue of data imbalance should be addressed. This can be achieved through focal loss, class-weighted log loss, or data augmentation. Additionally, it might be beneficial to use an active learning approach and annotate only those messages that the model is unsure about. Another approach is to combine multiple classifiers, such as a general-purpose classifier and specialized classifiers for different disaster types.

Second, the performance for rare categories should be improved. One possible approach to do that is to create hierarchical classifications, where messages are classified into broad categories first and then into subcategories. Another approach is to use few-shot and meta-learning methods that allow training classifiers with small amounts of training data.

Third, the model should be optimized for production, which requires reducing the model's size and latency. This can be achieved through model compression, such as pruning and quantization. Another approach is to use ONNX runtimes for CPU inference. Finally, a hybrid approach can be used, where TF-IDF based models perform fast first-pass screening, while more complex models such as transformers are used for specialized tasks.

Finally, it will be important to expand the domain of the model to include other languages and disaster types. This will ensure that the model generalizes well across different use cases and provides reasonably good performance even when the number of training samples is limited.

5.4 Summary

This project demonstrates the ability of machine learning algorithms to automatically classify disaster response messages. Among the algorithms evaluated, the transformer achieved the best results on all tasks. It can effectively predict urgency levels, detect disaster-related messages, and identify essential response categories.

The proposed approach provides a solution for automatically screening disaster-related messages for essential information that can guide the emergency response efforts. The transformer-based model achieved extremely high performance on all tasks, including an accuracy of 85.79% and an F1-Macro score of 82.28% on the urgency classification task, an F1-Score of 84.91% on the binary classification task, and an F1-Micro score of 75.02% on the multi-label classification task.

Despite the high performance, several limitations should be considered, including the effects of data imbalance, synthetic labels, rare categories, computational cost, and domain specificity. The performance of the models can be further improved through data augmentation, active learning, focal loss, and the use of specialized classifiers. Additionally, optimization techniques such as model compression and hybrid approaches can be used to reduce the model size and latency. Finally, future research should focus on evaluating the proposed approach on multi-lingual data sets and diverse disaster types.

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